dinosaurs. But there must be a point of diminishing returns: Clearly, some flux is too high for the continuance of any civilization.

One consequence of this train of argument is that, even if civilizations commonly arise on planets throughout the Galaxy, few of them will be both long-lived and nontechnological. Since hazards from asteroids and comets must apply to inhabited planets all over the Galaxy, if there are such, intelligent beings everywhere will have to unify their home worlds politically, leave their planets, and move small nearby worlds around. Their eventual choice, as ours, is spaceflight or extinction.

---

*The Outer Space Treaty, adhered to both by the United States and Russia, prohibits weapons of mass destruction in "outer space." Asteroid deflection technology constitutes just such a weapon—indeed, the most powerful weapon of mass destruction ever devised. Those interested in developing asteroid deflection technology will want to have the treaty revised. But even with no revision, were a large asteroid to be discovered on impact trajectory with the Earth, presumably no one's hand would be stayed by the niceties of international diplomacy. There is a danger, though, that relaxing prohibitions on such weapons in space might make us less attentive about the positioning of warheads for offensive purposes in space.

*What should we call this world? Naming it after the Greek Fates or Furies or Nemesis seems inappropriate, because whether it misses or hits the Earth is entirely in our hands. If we leave it alone, it misses. If we push it cleverly and precisely, it hits. Maybe we should call it "Eight Ball."

*There is of course a wide range of other problems brought on by the devastatingly powerful technology we've recently invented. But in most cases they're not Camarinan disasters—damned if you do and damned if you don't. Instead they're dilemmas of wisdom or timing—for example, the wrong refrigerant or refrigeration physics out of many possible alternatives.